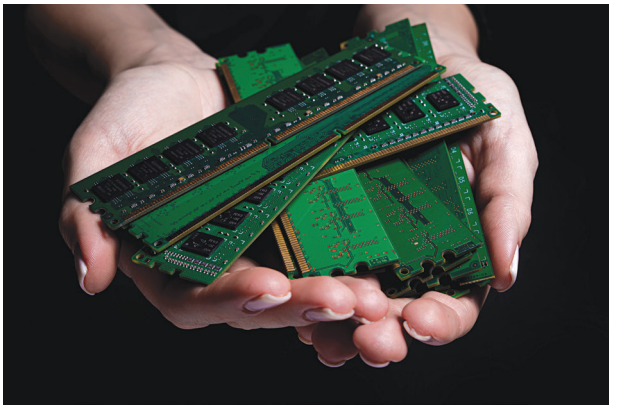
does not fit in its local memory, the runtime has to move some work elsewhere or split the model. That is why a slower GPU with more VRAM can sometimes be more useful for local AI than a faster gaming GPU with a smaller framebuffer.

Current NVIDIA laptop GPUs illustrate the spread. RTX 5050 and 5060 Laptop GPUs carry 8GB of GDDR7; the RTX 5070 can be configured with 8GB or 12GB; RTX 5070 Ti offers 12GB; RTX 5080 has 16GB; and RTX 5090 Laptop reaches 24GB. For local inference, those VRAM differences can determine model choice more directly than the gaming performance hierarchy.

Roughly speaking, 8GB VRAM is a useful entry point for smaller quantised models. Twelve to 16GB opens a substantially broader range. Twenty-four gigabytes makes many serious local workloads practical, though enormous models can still exceed it.

And remember the extra memory needed for context. Ollama's current defaults illustrate why VRAM changes behaviour: systems below 24GiB VRAM default to a 4K context, 24-48GiB systems to 32K, and systems with at least 48GiB to 256K. Ollama explicitly notes that increasing context increases memory requirements.



RAM: the capacity that saves you when VRAM cannot

System RAM matters even if you own a discrete GPU.

If a model fits entirely in VRAM, you can generally keep most of its heavy computation on the GPU. If it does not, runtimes such as llama.cpp can perform hybrid CPU-GPU inference, offloading some model layers to the GPU while keeping the rest in system memory. Ollama can likewise show models divided between CPU and GPU memory.

This makes a machine with 32GB or 64GB of system RAM surprisingly capable. You may be able to load a model far larger than your GPU alone could hold.

There is a catch: speed.

Moving data between system RAM and GPU memory across the system interconnect is not the same as keeping everything in fast local VRAM. CPU-offloaded models can work perfectly well, but the experience may be significantly slower. Mistral's own local-model guidance notes that CPU offload can make large models possible on machines with sufficient RAM, while warning that it is substantially slower.

For local AI, 16GB should increasingly be considered an entry-level capacity rather than a luxurious amount. Thirty-two gigabytes gives Windows, your runtime and the model much more breathing room. Sixty-four gigabytes becomes genuinely useful if you enjoy experimenting with larger quantised models.

STEP-BY-STEP: AUDIT THE PC YOU ALREADY OWN

Do this before shopping for hardware.

- **Step 1: Check system RAM.** On Windows, open Task Manager > Performance > Memory. On macOS, use Apple menu > About This Mac or System Information.
- **Step 2: Identify your GPU and VRAM.** In Windows Task Manager, open Performance > GPU and note Dedicated GPU Memory. Mac users should record unified-memory capacity instead.
- **Step 3: Record your CPU.** Note the exact processor model, not merely "Core i7" or "Ryzen 7".
- **Step 4: Check free SSD space.** Ollama warns that model libraries can consume tens or hundreds of gigabytes. Leave room for multiple quantisations and models.
- **Step 5: Check runtime compatibility.** Before assuming your GPU will accelerate anything, confirm support in LM Studio, Ollama or llama.cpp.
- **Step 6: Write down your baseline**: RAM, VRAM/unified memory, CPU, operating system and free storage. This becomes your local-AI specification sheet.

Unified memory: why Macs behave differently

Apple Silicon complicates the usual RAM-versus-VRAM discussion because CPU and GPU use a shared unified-memory pool. Instead of buying a machine with, say, 32GB system RAM plus 8GB dedicated VRAM, an Apple Silicon Mac may provide one memory pool that both CPU and GPU can access.

This can be extremely useful for local AI. A Mac with a large unified-memory configuration can make much more memory visible to GPU-accelerated workloads than a conventional laptop whose discrete GPU is limited to 8GB or 12GB.

Apple's MLX framework is designed specifically for machine learning on Apple Silicon, and llama.cpp treats Apple Silicon